

Design of a ventilation system for an office building with respect to energy efficiency and indoor climate

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SUMMARY

The task of the assignment is to design a complete ventilation system for a one-story office building in Gothenburg. The goal is to achieve an efficient ventilation system that is used for cooling and contributes to and reaches the requirements for thermal comfort.

The system integrates energy-efficient air handling units (AHUs) with heat recovery, optimized ductwork to minimize pressure losses, and strategically placed diffusers to ensure draft-free air distribution. Computational tools (Rhino, Grasshopper, Revit) were used to simulate energy performance and model airflow and thermal loads, and balancing sustainability with Gothenburg's climatic demands. The report outlines methodology, results, and system performance in achieving regulatory compliance, energy efficiency, and occupant comfort.

1 INTRODUCTION

The task is to create an indoor environment that contributes to a comfortable environment that does not endanger people's health. The ventilation system is supposed to keep the indoor environment constant, even if the outdoor temperature decreases drastically or if the amount of people increases/decreases. Clean air should be applied to the rooms where people spend most of their time [1].

Indoor air quality (IAQ) and thermal comfort are vital for occupant health, particularly as individuals spend 90% of their time indoors [1]. Gothenburg's temperate coastal climate—cold, humid winters, and mild summers—necessitates a ventilation system addressing heating, cooling, and humidity control while prioritizing energy efficiency. This study designs a constant air volume (CAV) mechanical ventilation system for a local office building, aligning with Boverkets Byggregler (BBR) standards [2].

2 METHODOLOGY

The designed ventilation system was assumed to be an all-air system for an open-area office (Figure 2.1). In this study, the ventilation system for supplying and exhausting air from the office area was designed. While another system handles air extraction from the toilets and kitchen. To avoid oversizing the system, the winter design airflow was assumed to be equal to the summer design airflow. The ventilation system is intended for cooling, with an auxiliary system assumed to cover the peak heating load during winter.

2.1 Energy Simulation

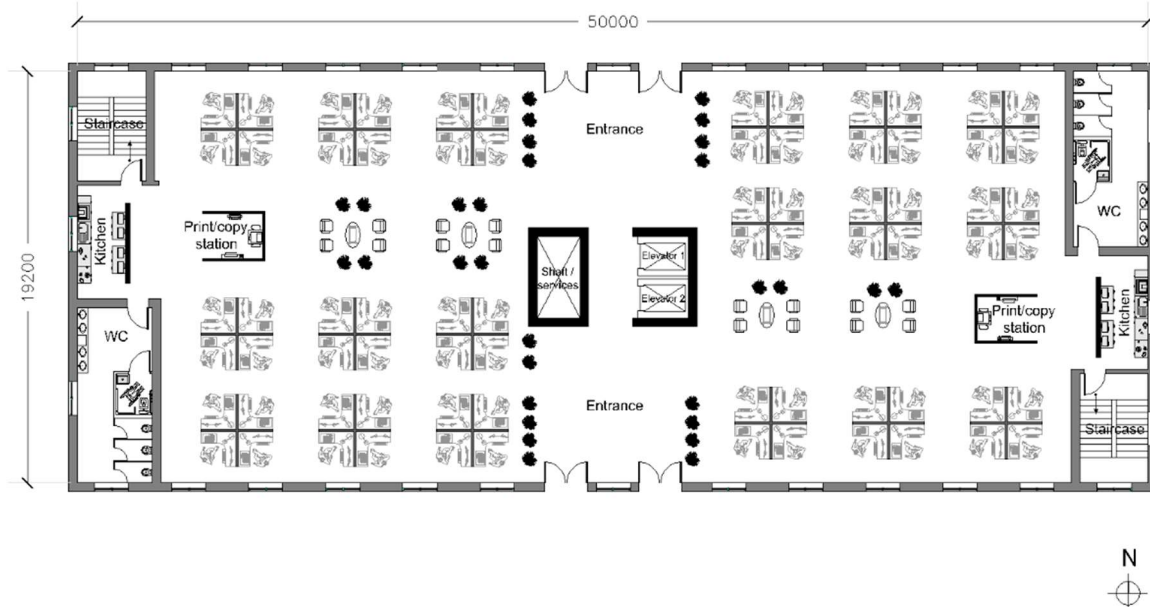


Figure 2.1 - floor layout of the open office

To design a CAV system, the airflow required for cooling during peak loads was calculated using an energy simulation performed with Rhino 8, Grasshopper, and ClimateStudio. Based on the information provided in the assignment, the building was in Gothenburg and was constructed with a wooden structure. Envelope properties (Table 1) and loads (Table 2) were assigned in the simulation, and to ensure a realistic scenario, a 15% thermal bridge was accounted for in all envelopes. This model was referred to as the Base case in this report.

Table 1- Envelopes' property

	External wall	Roof	Foundation	Window (south & north)	Window (east & west)	Doors
U-Value W/(m ² .K)	0.15	0.11	0.12	1.4	1.5	1.8

The office building is expected to meet the thermal comfort requirements. According to FEBY18 [3], the indoor operative temperature during the period from April to September should not exceed 26°C for more than 10% of the time. To achieve this requirement, passive strategies were applied to the building to mitigate overheating rather than relying excessively on the cooling system.

Table 2 - Inputs used for energy simulation

	Base case	Improved Case
People (P/m ²)	0.075	0.075
Lighting (W/m ²)	10	7
Equipment (W/m ²)	15	15
Min. airflow per person (l/s per person)	7	7
Min. airflow per area (l/s/ m ²)	0.35	0.35
Infiltration (ACH)	0.2	0.2
Heating setpoint (°C)	21	21
cooling setpoint (°C)	26	26
Heat recovery	none	sensible

For the simulation, the maximum supply air temperature was set to 20°C in winter, while the minimum supply air temperature was set to 18°C in summer. This corresponds to a temperature difference of 8 K between the room and supply air temperature for summer. The EPW weather data for Gothenburg provided by Ladybug tools was used for simulation [4]. The total airflow required during the peak load in summer was calculated with the use of Equation 1 and Equation 2. The results for energy demand, cooling peak load, and airflow for Base and improved case are shown in Table 4.

$$m = \frac{Q}{C_p \cdot \Delta T} \quad \text{Equation 1}$$

Where m is the mass flow rate (kg/s), Q is the room peak load (kW), C_p is the specific heat capacity of air (J/(kg°C)), ΔT is the temperature difference between inlet and outlet air (°C).

$$V = \frac{m}{\rho} \quad \text{Equation 2}$$

Where V is airflow (m³/s), m is mass flow rate (kg/s), ρ is air density (kg/m³).

2.2 Ducts System

The strategy when designing the ventilation system aimed to ensure that the employees receive an appropriate airflow in the area where they spend the majority of their workday. The strategy is also included to ensure that the supply and exhaust diffusers are not placed too close to avoid draught or short circuits (when the supply air gets exhausted too fast). The supply diffusers were therefore placed on top of the desks and close to the hallway, and the exhaust diffusers were placed at a marginal distance from the supply diffusers close to the exterior walls, kitchen, and printing area.

The assignment also highlighted that if the office wishes to reorganize the placement of furniture, it would be possible independently of the ventilation system's design. Consequently, the supply and exhaust diffusers were evenly distributed throughout the open office area to accommodate potential future reorganization. Additionally, tees and removable caps were installed beside elbows where diffusers were omitted due to furniture placement, allowing for future modifications if needed.

In the planning of the ventilation system circular ducts were used. Circular ducts are cheaper, more airtight, and easier to manage than rectangular ducts. The sheet is thinner because the shape is stiffer, in addition, the cross-sectional area is more effective for airflow [1].

A chart was provided by the assignment for circular ducts where the dimensions of the ducts were chosen depending on the friction loss and airflow. The airflow of each duct was calculated based on the number of diffusers in each branch and the required airflow for each. Based on the assignment requirement, the friction loss for all ducts should be lower than 1 Pa/m. So, based on the airflow and using the chart, optimized duct size was selected (Figure 3.2). The idea was to choose as few different ducts size as possible to not get a high-pressure drop; while ensuring a low-velocity system. Since the low-velocity system reduces friction loss, noise level and lower fan power. On the other hand, oversizing the ducts means usage of more material and higher initial cost. So, the decision was made in a way to make a balance between air velocity and material efficiency.

After sizing the ducts, friction loss and air velocity for each part were analyzed using the above-mentioned chart. Friction loss for fittings was calculated using the equivalent length method, where tees, elbows, Y-junctions, and transitions were considered to have equivalent losses of 15, 10, 8, and 2 meters of duct length, respectively. The final friction loss for each diffuser was calculated by this method in an Excel sheet and critical paths for both supply and exhaust air were identified.

Both supply and exhaust ducts were connected to the air handling unit through the main shaft in the building. This made the possibility of planning ducts relatively symmetrical to make balanced and smooth air flow and lowering the friction loss.

2.3 Diffusers

The selection of diffusers was made in Swegon Software [5]. The strategy for this part was to reduce any kind of draught, maintain in the acceptable range for carbon dioxide content, and sound from the diffusers. When dimensioning the open office area, the office was divided into 2 zones (half of the office area) when simulations were made in Swegon. The office model had

a height of 2.8 meters, with 2 meters designated as the occupied zone height. The supply air temperature was set to 18 °C. Table 3 illustrates the requirements for selecting the diffusers.

Table 3 - Required values to fulfill for the diffuser and the office area

Parameters	Values
Sound level	< 30 dB
Carbon dioxide content (CO ₂)	1000 PPM
Room temperature	21 °C
Highest air speed in occupancy zone (cooling)	0.25 m/s
Max. pressure drop	50 Pa

2.4 Air Handling Unit (AHU)

The positioning of the AHU and distribution shafts is important in HVAC design. To balance accessibility for maintenance and noise reduction, AHUs are commonly installed on rooftops or within designated mechanical rooms [6]. This approach not only streamlines servicing but also reduces noise produced in occupied areas. Distribution shafts, acting as central conduits for airflow, are ideally placed near the building's core.

The AHU sizing process involves determining the required airflow for the building and ensuring that the selected air handling unit meets the desired air velocity for optimal performance. The study referenced the Fläkt Woods (EU2000) table (see Figure 01 1 in Appendix 01) to determine the AHU's size and efficiency. The chart is used to compare various AHU sizes, each with corresponding front areas. By dividing the required airflow by the front area of each unit, the resulting air velocity is calculated (Equation 3). The goal is to choose a unit that maintains an air velocity below a set limit, which helps to maintain noise and pressure losses. Additionally, the selected AHU should have sufficient fan capacity to handle the airflow while optimizing energy efficiency and system performance. In HVAC design, typical duct velocities range from 2 to 5 m/s for comfort applications and higher for industrial settings [7].

$$Velocity = \frac{\dot{V}}{A} \quad \text{Equation 3}$$

Where *Velocity* is air velocity (m/s), $\dot{V}$ is the airflow rate (m³/s), and *A* is the sectional area (m²)

The heating and cooling coil capacities were dimensioned using Equation 4 and Mollier diagram analysis, which tracked air path processes from outdoor air (OA) to supply air (SA). For the Outdoor design conditions extreme temperatures were extracted from the Gothenburg EPW weather file. Outdoor air conditions during peak cooling loads were also analyzed. While the peak cooling demand occurred in the morning hours, the maximum outdoor temperature showed a comparable cooling load due to higher solar gains. This difference arises from pre-cooling requirements before occupancy. As a result, for cooling size calculation the most severe outdoor condition were applied to calculate maximum enthalpy differentials. For the heating coil also heat recovery (latent and sensible) were considered. To avoid meeting the dew point in heat recovery system 2 °C preheating was used.

$$\dot{Q} = \dot{V} \cdot \rho \cdot (h_2 - h_1) \quad \text{Equation 4}$$

Where $\dot{Q}$ is the power of the fan (kW), $\dot{V}$ is the airflow (m^3/s), ρ is the density of air (kg/m^3), h_2 is the coil-off condition (kJ/kg), and h_1 is the coil-on condition (kJ/kg).

Equation 5 was used to calculate the Total Fan Power Input. To assess total pressure drops across various AHU components, the table from “*Achieving the Desired Indoor Climate*” was utilized [8] (see Figure 01 2 in Appendix 01). Since the designed system is a low velocity, the preferable total values were used for calculation. Pressure drops for critical paths in both supply and exhaust systems were added to the pressure drop calculation. The whole calculation for pressure drops can be seen in Appendix 03. Centrifugal fan with 0.65 efficiency was selected for the project.

$$\dot{W}_t = \frac{\dot{V} \cdot \Delta p_{tot}}{\eta_{tot}} \quad \text{Equation 5}$$

Where $\dot{W}_t$ is the electric power of the fan (kW), $\dot{V}$ is the airflow rate through fan (m^3/s), Δp_{tot} is the total pressure rise of the fan (kPa), and η_{tot} is the efficiency of the fan.

Specific Fan Power (SFP) is an efficient indicator that measures the energy required to move air through a system. It is calculated by Equation 6.

$$SFP = \frac{\dot{W}_t}{\dot{V}} \quad \text{Equation 6}$$

Where SFP is the specific Fan Power ($\text{kW}/(\text{m}^3/\text{s})$), $\dot{W}_t$ is the electric fan power (kW), and $\dot{V}$ is the airflow rate (m^3/s)

3 RESULTS AND DISCUSSION

3.1 Energy and Airflow

The following improvement strategies were implemented in the base case to reach the thermal comfort requirement:

- Horizontal fixed shading devices (100 cm depth) on the south-, east-, and west-facing windows
- Vertical fixed shading devices (30 cm depth) on south-, east-, and west-facing windows
- Reduction of the Solar Heat Gain Coefficient (SHGC) of glazing from 0.65 to 0.3 to minimize solar heat entry during summer
- Reduction of lighting power density from 10 W/m^2 to 7 W/m^2 by using energy-efficient lighting while maintaining equivalent illuminance
- Optimization of the cooling system schedule to activate two hours before working hours to reduce peak loads
- Addition of sensible heat recovery to the ventilation system (80%)

By applying improving steps, the percentage of hours in which indoor operative temperature (April – September) exceeds 26°C decreased from 51% to 10%. As shown in Table 4 addressing the overheating issue, also led to a reduction in overall energy usage and peak loads for cooling and heating. Lower peak loads directly reduced the required airflow, allowing for smaller duct sizes and lower fan power in the ventilation system. Since the ventilation system must be

designed to accommodate cooling demands, the airflow required at peak cooling load is considered in the following design steps.

Table 4 - Energy peak loads and airflow required for cooling for Base and Improved case

	Base Case	Improved case
Energy Use Intensity (kWh/m ² /year)	95	68.9
Cooling peak load (kWh)	34.08	18.22
Airflow(m ³ /s)	35.50	18.98

3.2 Duct system

The Duct system was designed as symmetrical as possible by a tree layout. To ensure sufficient fresh air for people one diffuser was placed on top of each working unit and sitting area. To prevent the distribution of pollutants, air diffusers were omitted in the printer area. Also, two diffusers were placed in the middle of commuting areas. Exhaust diffusers were placed close to the exterior walls and far from the occupant area. Two main ducts for the exhaust system were considered in the main shaft to avoid a cross between supply and exhaust ducts.

While minimizing the bends and length is important, to avoid cross-fitting each diffuser connected to the branch with a separate feeding from the side. The length of the feedings is 3 times the duct size. Tees and removable caps were installed at the end of the branch to allow for future modifications and easy access for maintenance tasks such as cleaning. Figure 3.1 illustrates the layout of ducts for the supply and exhaust system.

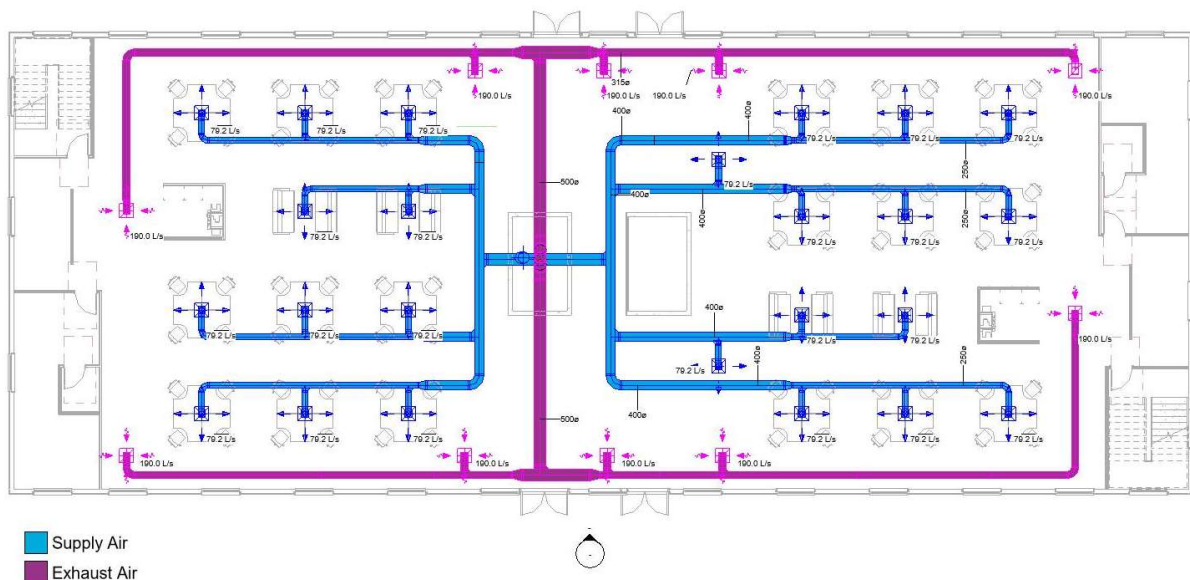


Figure 3.1- Duct layout for supply and exhaust system with 24 supply diffusers and 10 exhaust diffusers

Friction loss and air velocity in each duct, extracted from the friction loss chart (Figure 3.2). Pressure drop for fittings was calculated with an equivalent length. Critical path identified based on the total length and total pressure drop for each diffuser. Appendix 03 shows the calculation of pressure drop in all diffusers and critical paths. Pressure drops were 68.5 Pa and 97.6 Pa for supply and exhaust critical paths, respectively.

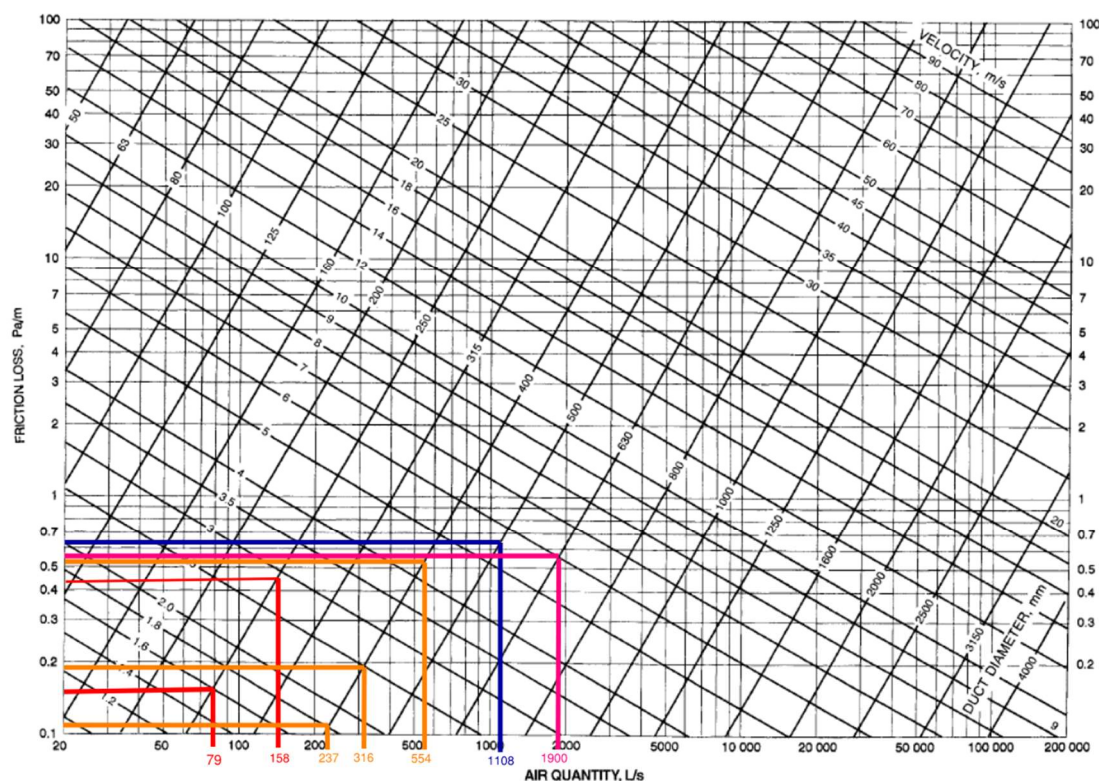


Figure 3.2- Friction loss for circular ducts based on airflow and duct size

3.3 Diffusers

The chosen supply diffuser was EAGLE CRa 315-600-RO + ALSd 250-315. This type of diffuser supported CAV ventilation systems. The same diffuser type was selected for supply and exhaust systems. Table 5 presents the parameters for the diffusers.



Figure 3.3 - Supply diffuser: EAGLE CRa 315-600-RO + ALSd 250-315 (Swegon n.d.).

Table 5 - Parameters and values for the supply diffuser (Swegon n.d.).

Parameters	supply diffusers	Exhaust diffusers
Sound	< 20 dB	24 dB
Airflow	79 l/s	190 l/s
Distribution pattern	Rotation (RO)	-
Pressure drops	5.2 Pa	50 Pa
Size	600 mm	600 mm
Neck size of the duct system	250 mm	315 mm
Commissioning box	1-step	1-step

Figure 3.4 shows the airflow pattern for cooling and heating for half of the open office building. The diffusers are placed to not cause any draught, that would happen if they were too close to one another.

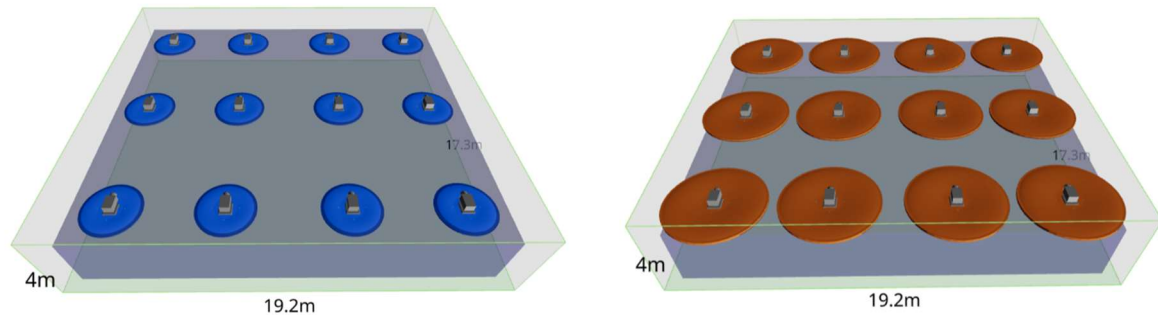


Figure 3.4 - Cooling (left) and heating (right) pattern for the office (Swegon n.d.).

3.4 Air handling unit

A total airflow of $1.9 \text{ m}^3/\text{s}$ was needed for the whole building. AHU size was selected to reach air velocity lower than 2.5 m/s . By dividing the required airflow by the front area of the AHU, air velocity can be calculated. Based on that AHU size 30 with max fan power of $3.5 \text{ m}^3/\text{s}$ was selected for the project. Using lower sizes was also possible regarding required airflow but not optimized considering the air velocity (Table 6). Final air velocity for the system was 2.35 m/s .

Table 6 AHU selection based of airflow and air velocity

AHU size	Fan max (m^3/s)	Required airflow (m^3/s)	Front area (m^2)	Air velocity (m/s)
21	2.4	1.9	0.54	3.52
22	3.2	1.9	0.72	2.64
30 (selected)	3.5	1.9	0.81	2.35

Based on the graph provided and considering the equal amount of supply and exhaust air ($1.9 \text{ m}^3/\text{s}$) the temperature efficiency for AHU size 30 was detected to 83% (Figure 3.5Figure).

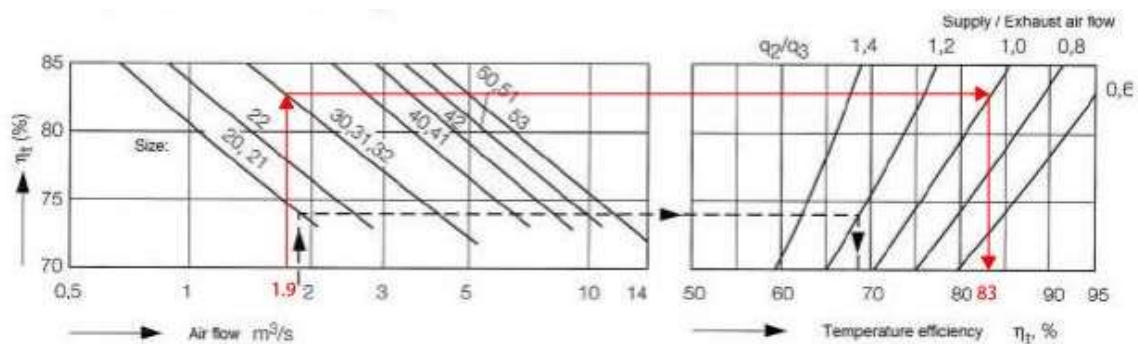


Figure 3.5 Temperature efficiency of the selected AHU

The power required for the heating and cooling coils was calculated using Mollier diagram. Climate data extracted from the Gothenburg EPW file indicated design conditions of -15.9°C (84% relative humidity) for the coldest day and 27.1°C (41% relative humidity) for the

warmest day of the year. A 1°C temperature rise was applied to outdoor air temperatures in both heating and cooling scenarios to account for heat losses in the ventilation pump (see Figure 04 1 and Figure 04 2 in Appendix 04)

For the heating system, the indoor design conditions were set to 21°C and 40% relative humidity. The outdoor air required heating to 20°C before entering the occupied space. To avoid condensation in heat exchanger 2 °C preheating considered before heat recovery. Using the heat recovery system's 83% efficiency, 83% of the heating demand was met by recovered heat, with the remaining provided by the heating coil. This calculation followed the enthalpy difference method on the Mollier diagram, resulting in 4.58 kW and 19.24 kW for the power of preheating and heating coil, respectively (Equation 7 & Equation 8).

For the cooling system, outdoor air was cooled to minimum supply temperature of 18°C. The process involved tracing the path on the Mollier diagram from outdoor conditions (27.1°C, 41% RH) to the apparatus dew point of 5°C. The cooling coil capacity was determined to be 32.83 kW, derived from the total enthalpy change between the outdoor air and minimum supply temperature (Equation 9).

$$Q_{preheating} = 1.9 \times 1.2 \times (13.79 - 11.78) = 4.58 \text{ kW} \quad \text{Equation 7}$$

$$Q_{heating} = 1.9 \times 1.2 \times (35.84 - 27.4) = 19.24 \text{ kW} \quad \text{Equation 8}$$

$$Q_{cooling} = 1.9 \times 1.2 \times (51.6 - 37.2) = 32.83 \text{ kW} \quad \text{Equation 9}$$

The specific fan power (SFP) was calculated as follows with provided values from Figure 01 2 and calculation of pressure drop for critical paths.

$$SFP = \frac{v \cdot \Delta p_{tot}}{\frac{n_{tot}}{v}} = \frac{\Delta p_{tot}}{n_{tot}} = \frac{816.1}{0.65} \approx 1.25 \text{ kW/m}^3/\text{s} \quad \text{Equation 10}$$

Efficient ventilation systems have a value between 1.0 – 2.0 kW/m³/s. Improvements could be made by reducing the longest critical path, the amount of pressure drops (elbows, tees, duct reducers etc.).

3.5 Discussion

- Since a completely symmetrical system was not feasible due to the furniture layout and shaft location, to maintain a consistent number of duct sizes and create a pattern for easier installation, oversized ducts were used in some branches.
- Utilizing the ceiling area to install the duct system (0.9 m) made it difficult to use diffuser models with top connections due to the limited space. As a result, diffusers with side connections were used, leading to longer ducts, more fittings, and increased friction loss.
- Fixed occupancy schedules and equipment loads (15 W/m²) may not accurately reflect real-world variations, which could influence peak load calculations.
- Pressure loss calculations were based on equivalent duct lengths for fittings, which simplifies turbulent airflow dynamics and may underestimate pressure drops.

- The assumed 80% sensible heat recovery efficiency is based on ideal conditions, without accounting for potential degradation over time.
- Further investigation into other AHU components, such as filters, dampers, humidifiers, and fire cells, should also be considered in the next steps.

4 CONCLUSIONS

This study connects theoretical modelling with practical HVAC design, offering a clear pathway to regulatory compliance, improved occupant comfort, and enhanced energy efficiency. While some simplifications were made, the proposed strategies provide a strong foundation for ongoing advancements in ventilation system design.

This study highlights how integrating passive design strategies with optimized mechanical systems can significantly improve thermal comfort and energy efficiency in office buildings. By reducing overheating hours, key measures such as shading devices, and pre-cooling schedules have proven highly effective in minimizing solar heat gain. Additionally, the 27% reduction in annual energy use intensity demonstrates both the economic and environmental advantages of this approach.

Throughout the design process, it became evident that the placement of diffusers and the layout of ductwork play an important role in determining system pressure drops. To minimize unnecessary pressure losses, it is essential to limit the number of dimension reductions in the ductwork. Additionally, diffuser placement significantly impacts thermal comfort for occupants. A well-designed system should ensure even air distribution across all workstations.

The primary objective of this project was to ensure that friction loss did not exceed 1 Pa/m when selecting duct sizes. As a result, velocity was not a primary focus, since the provided values for calculating pressure drop did not account for it. However, for larger ventilation systems in the future, airflow velocity could become a key factor in efficiently distributing air at higher speeds. In such cases, the approach to calculating pressure drop would need to be adjusted accordingly, leading to different design outcomes.

5 ACKNOWLEDGEMENT & CONTRIBUTION

Group 12 would like to thank Eszter Krivácsy and Ana Sofia Compean Becerra for their help with the process of the AHU design.

Lukas Rosberg contributed to the diffuser selection and airflow, friction loss calculation, AHU selection, ducting, and building modeling in Revit.

Vinay Dilip Jayaselan contributed to the energy simulation, diffuser selection, coil sizing with Mollier chart, ducting, and duct system modeling in Revit.

Sakineh J. Z. Afshari contributed to the energy simulation, ducting, friction loss calculation, AHU selection, and coils sizing with Mollier chart.

6 DECLARATION OF AI-ASSISTANT TECHNOLOGIES

During the preparation of this work the authors used ChatGPT in order to identify the source of information, grammar check, and improvement in readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the report.

7 REFERENCES

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Appendix 01

SIZE	AIR FLOW m ³ /s				DIMENSIONS			
	Fan max m ³ /s	Air velocity			Front area m ²	Cross-section area		
		2.5 m/s	3.0 m/s	4.0 m/s		Outer Height mm	Width mm	Inner m ²
11	0.9	0.5	0.5	0.7	0.18	474	804	0.26
20	1.6	0.9	1.1	1.4	0.36	774	804	0.47
21	2.4	1.4	1.6	2.2	0.54	774	1104	0.67
22	3.2	1.8	2.2	2.9	0.72	774	1404	0.87
30	3.5	2.0	2.4	3.2	0.81	1074	1104	0.97
31	4.8	2.7	3.2	4.3	1.08	1074	1404	1.26
32	6.0	3.4	4.1	5.4	1.35	1074	1704	1.55

Figure 01 1 - Table from Fläkt Woods, example of quick choice for AHUs

Component	Supply air Pa	Exhaust air Pa	Total Pa	Preferable total Pa
Dampers	10	10	20	20
Filters	180	160	340	250
Air-to-air heat exchanger	150	150	300	200
Heating coil	40		40	20
Cooling coil	100		100	60
Sound attenuators	50	50	100	50
Fan system effects	100	100	200	50

Figure 01 2 - Values for pressure drops across various AHU components, table from "Achieving the Desired Indoor Climate"

Appendix 02 Duct system and connection to AHU

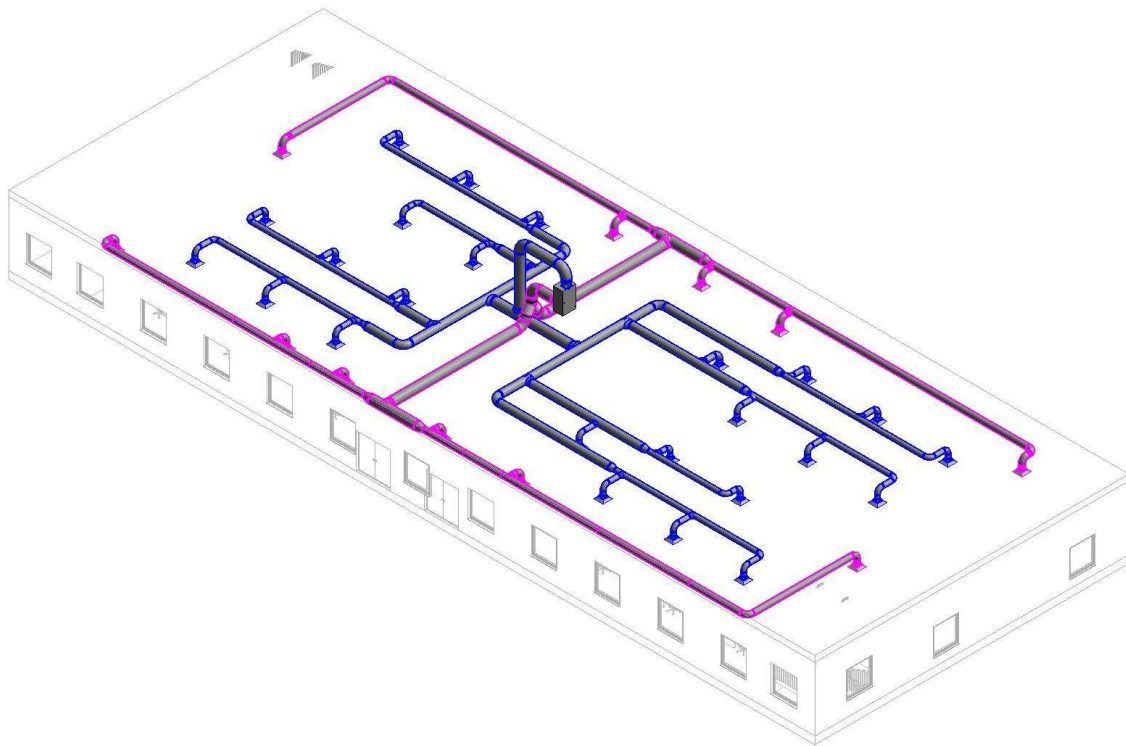


Figure 02 1 - 3D view of the duct systems

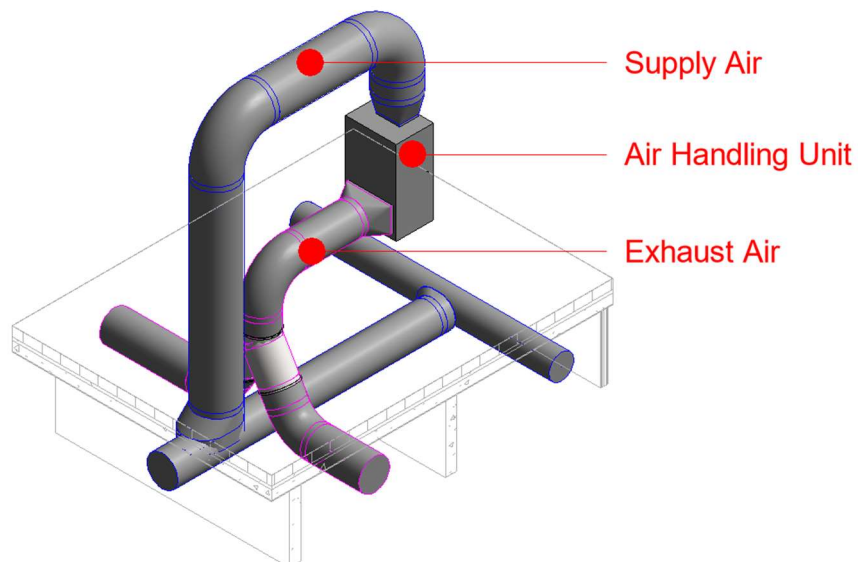


Figure 02 2 - Connection of exhaust and supply ducts to the AHU

Appendix 03 Pressure drop calculation for the duct system

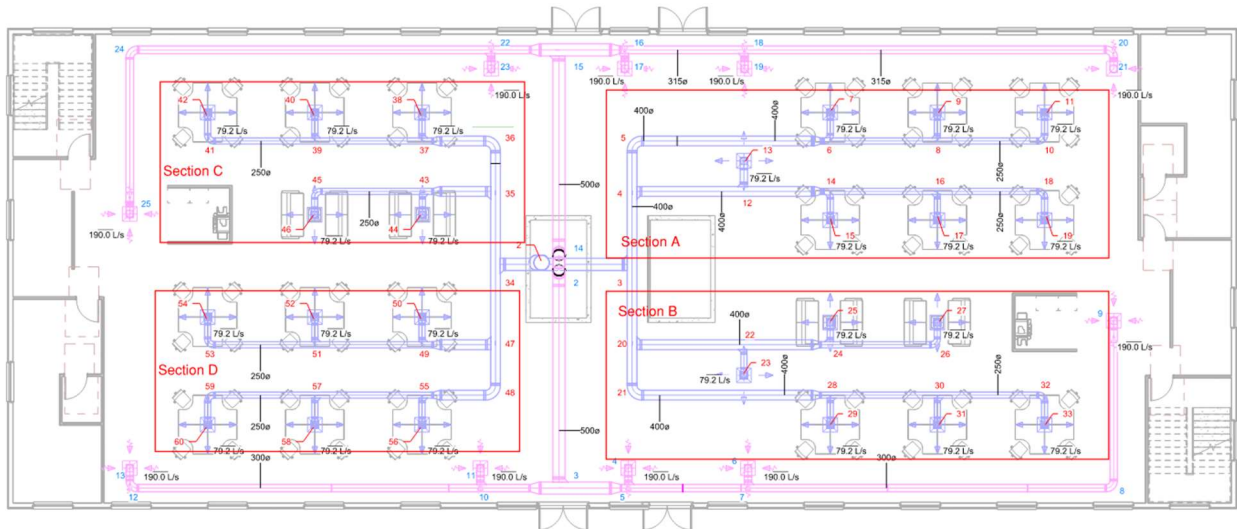


Figure 03 1- Drawing of the Duct systems and diffusers

Section	Flow rate	Velocity	Diameter	$\Delta P/L$	Length	Duct loss	Fitting type	Fitting loss	Total pressure loss
A	l/s	m/s	mm	Pa/m	m	Pa	-	Pa	Pa
0-1 (horizontal from AHU)	1900	6	630	0.54	1.75	0.95	elbow+ elbow	10.8	11.7
1-2 (vertical throught the shaft)	1900	6	630	0.54	2.90	1.57	tee + transition	9.18	10.7
2-3	1108	5.5	500	0.62	3.50	2.17	tee + transition	10.54	12.7
3-4	554.19	4.3	400	0.54	2.54	1.37	tee	8.1	9.5
4-5	237.5	1.85	400	0.11	1.16	0.13	elbow	1.1	1.2
5-6	237.5	1.85	400	0.11	7.30	0.80	tee + transition	1.87	2.7
6-8	158.3	3	250	0.49	4.14	2.03	tee	7.35	9.4
8-10	79.17	1.65	250	0.17	4.14	0.70	elbow	1.7	2.4
10-11	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:4:5:6:8:10:11									66.4
6-7	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:4:5:6:7									54.6
8-9	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:4:5:6:8:9									64.0
10-11	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:4:5:6:8:10:11									66.4
4-12	316.68	2.6	400	0.18	4.29	0.77	tee + transition	3.06	3.8
12-13	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:4:12:13									54.5
12-14	237.5	1.85	250	0.11	3.27	0.36	tee + transition	1.87	2.2
14-15	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:4:12:14:15									56.8
14-16	158.34	3	250	0.49	4.14	2.03	tee	7.35	9.4
16-17	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:4:12:14:16:17									66.1
16-18	79.17	1.65	250	0.17	4.14	0.70	Eelbow	1.7	2.4
18-19	79.17	1.65	250	0.17	0.75	0.1275	diffuser	5.9	6.03
0:1:2:3:4:12:14:16:18:19									68.54

Figure 03 2 - Pressure drop calculation for diffusers - Supply ducts - Section A

Section	Flow rate	Velocity	Diameter	$\Delta P/L$	Length	Duct loss	Fitting type	Fitting loss	Total pressure loss
B	l/s	m/s	mm	Pa/m	m	Pa	-	Pa	Pa
0-1 (horizontal from AHU)	1900	6	630	0.54	1.75	0.95	elbow+ elbow	10.8	11.7
1-2 (vertical throgh the shaft)	1900	6	630	0.54	2.90	1.57	tee + transition	9.18	10.7
2-3	1108	5.5	500	0.62	3.50	2.17	tee + transition	10.54	12.7
3-20	475.02	3.75	400	0.4	2.54	1.02	tee	6	7.0
20-21	237.5	1.85	400	0.11	1.16	0.13	elbow	1.1	1.2
21-28	237.5	1.85	400	0.11	7.30	0.80	tee + transition	1.87	2.7
28-30	158.3	3	250	0.49	4.14	2.03	tee	7.35	9.4
30-32	79.17	1.65	250	0.17	4.14	0.70	elbow	1.7	2.4
32-33	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:20:21:28:30:32:33									63.9
28-29	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:20:21:28:29									52.1
30-31	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:20:21:28:30:31									61.5
32-33	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:20:21:28:30:32:33									63.9
20-22	237.51	1.85	400	0.11	4.29	0.47	tee + transition	1.87	2.3
22-23	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:20:22:23									50.6
22-24	158.3	3	250	0.49	3.27	1.60	tee + transition	8.33	9.9
24-25	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:20:22:24:25									60.5
24-26	79.17	1.65	250	0.17	4.14	0.70	tee	2.55	3.3
26-27	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:3:20:22:24:26:27									63.8

Figure 03 3 - Pressure drop calculation for diffusers - Supply ducts - Section B

Section	Flow rate	Velocity	Diameter	$\Delta P/L$	Length	Duct loss	Fitting type	Fitting loss	Total pressure loss
C	l/s	m/s	mm	Pa/m	m	Pa	-	Pa	Pa
0-1 (horizontal from AHU)	1900	6	630	0.54	1.75	0.95	elbow+ elbow	10.8	11.7
1-2 (vertical throgh the shaft)	1900	6	630	0.54	2.90	1.57	tee + transition	9.18	10.7
2-34	1108	5.5	500	0.62	1.75	1.09	tee + transition	10.54	11.6
34-35	395.85	3.25	400	0.3	2.54	0.76	tee	4.5	5.3
35-36	237.5	1.85	400	0.11	1.16	0.13	elbow	1.1	1.2
36-37	237.5	1.85	400	0.11	2.70	0.30	tee + transition	1.87	2.2
37-39	158.3	3	250	0.49	4.14	2.03	tee	7.35	9.4
39-41	79.17	1.65	250	0.17	4.14	0.70	elbow	1.7	2.4
41-42	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:35:36:37:39:41:42									60.6
37-38	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:35:36:37:38									48.8
39-40	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:35:36:37:39:40									58.2
35-43	158.34	1.65	250	0.17	2.70	0.46	tee + transition	2.89	3.3
43-44	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:35:43:44									48.8
43-45	79.7	1.65	250	0.17	4.14	0.70	tee + transition	2.89	3.6
45-46	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:35:45:46									52.3

Figure 03 4 - Pressure drop calculation for diffusers - Supply ducts - Section C

Section	Flow rate	Velocity	Diameter	$\Delta P/L$	Length	Duct loss	Fitting type	Fitting loss	Total pressure loss
D	l/s	m/s	mm	Pa/m	m	Pa	-	Pa	Pa
0-1 (horizontal from AHU)	1900	6	630	0.54	1.75	0.95	elbow+ elbow	10.8	11.7
1-2 (vertical throgh the shaft)	1900	6	630	0.54	2.90	1.57	tee + transition	9.18	10.7
2-34	1108	5.5	500	0.62	1.75	1.09	tee + transition	10.54	11.6
34-47	475.02	3.75	400	0.4	2.54	1.02	tee	6	7.0
47-48	237.5	1.85	400	0.11	1.16	0.13	elbow	1.1	1.2
48-55	237.5	1.85	400	0.11	2.70	0.30	tee + transition	1.87	2.2
55-57	158.3	3	250	0.49	4.14	2.03	tee	7.35	9.4
57-59	79.17	1.65	250	0.17	4.14	0.70	elbow	1.7	2.4
59-60	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:47:48:55:57:59:60									62.3
55-56	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:47:48:55:56									50.6
57-58	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:47:48:55:57:58									59.9
47-49	237.51	1.85	400	0.11	2.70	0.30	tee + transition	1.87	2.2
49-50	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:47:49:50									64.5
49-51	158.34	1.65	250	0.17	2.70	0.46	tee + transition	2.89	3.3
51-52	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:47:49:51:52									52.7
51-53	79.7	1.65	250	0.17	4.14	0.70	tee + transition	2.89	3.6
53-54	79.17	1.65	250	0.17	0.75	0.13	diffuser	5.9	6.0
0:1:2:34:47:49:51:53:54									56.3

Figure 03 5 - Pressure drop calculation for diffusers - Supply ducts - Section D

Section	Flow rate	Velocity	Diameter	$\Delta P/L$	Length	Duct loss	Fitting type	Fitting loss	Total pressure loss
Exhaust	l/s	m/s	mm	Pa/m	m	Pa	-	Pa	Pa
9-8	190	2.15	315	0.24	5.90	1.42	diffuser+ elbow	52.4	53.8
8-6	190	2.15	315	0.24	14.20	3.41	tee	3.6	7.0
6-4	380	4.8	315	0.85	4.60	3.91	tee + transition	14.45	18.4
4-3	570	2.8	500	0.18	2.96	0.53	tee	2.7	3.2
3-2	950	4.8	500	0.48	7.50	3.60	elbow+Yjunction+transition	9.6	13.2
2-1	1900	6	630	0.55	1.54	0.85	transition	1.1	1.9
9:8:6:4:3:2:1									97.6
7-6	190.00	2.15	315	0.24	0.70	0.17	tee + diffuser	53.6	53.8
7:6:4:3:2:1									90.51
5-4	190.00	2.15	315	0.24	0.70	0.17	tee + diffuser	52.4	52.6
5:4:3:2:1									70.95
13-12	190	2.15	315	0.24	0.7	0.17	elbow + diffuser	52.4	52.6
12-10	190	2.15	315	0.24	14.27	3.42	tee + transition	4.08	7.50
10-3	380	3	500	0.27	1.85	0.50	tee	4.05	4.5
13:12:10:3:2:1									79.77
11-10	190	2.15	315	0.24	1.85	0.44	diffuser + tee	53.6	54.0
11:10:3:2:1									73.74

Figure 03 6 - Pressure drop calculation for diffusers - Exhaust system

Appendix 04 Dimensioning Heating and Cooling coils by Mollier chart

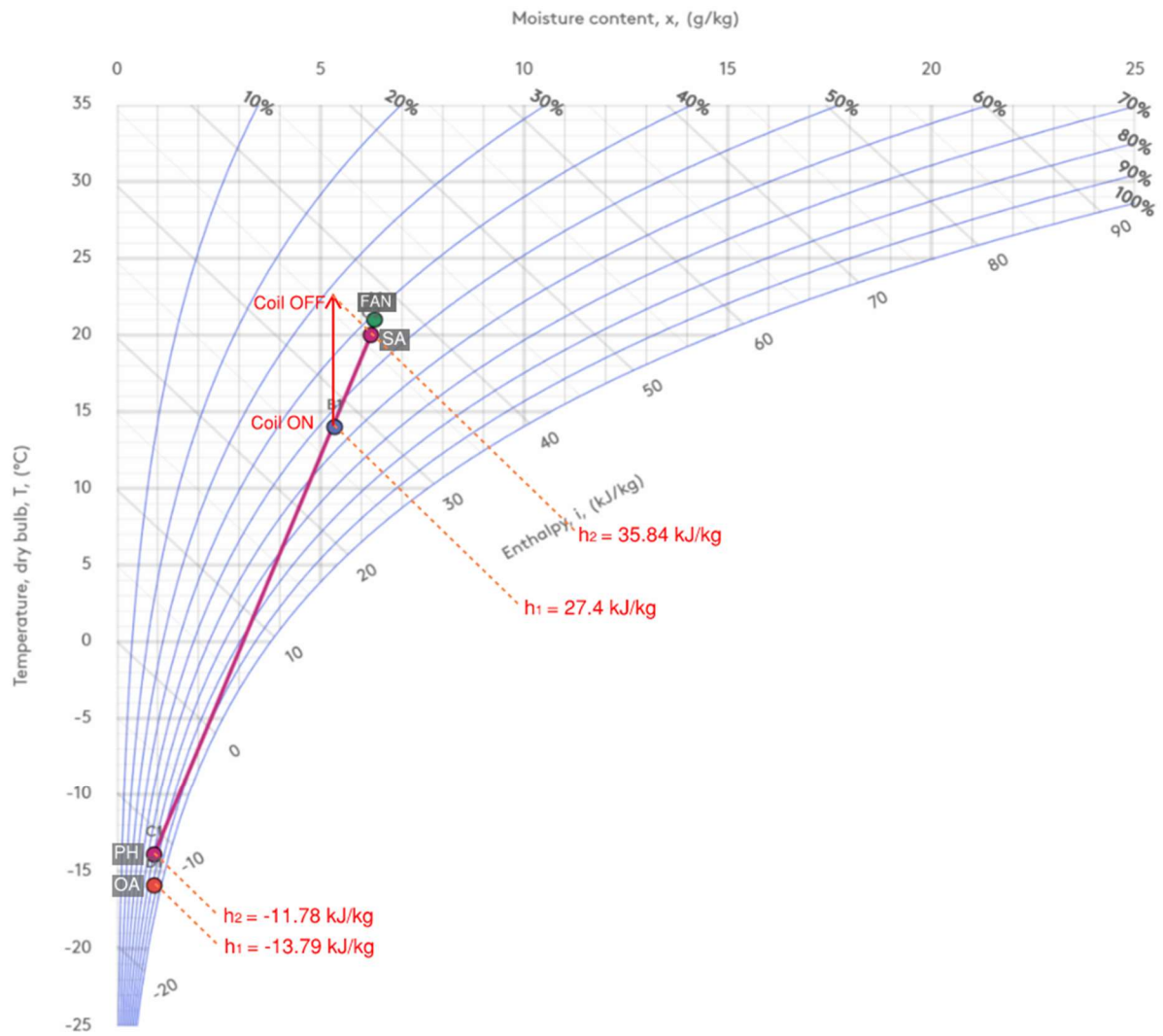


Figure 04 1 - AHU heating coil dimensioning showing Outdoor air (OA) pre-heated to (PH) and again heated from Coil-on condition to Coil-off condition (SA)

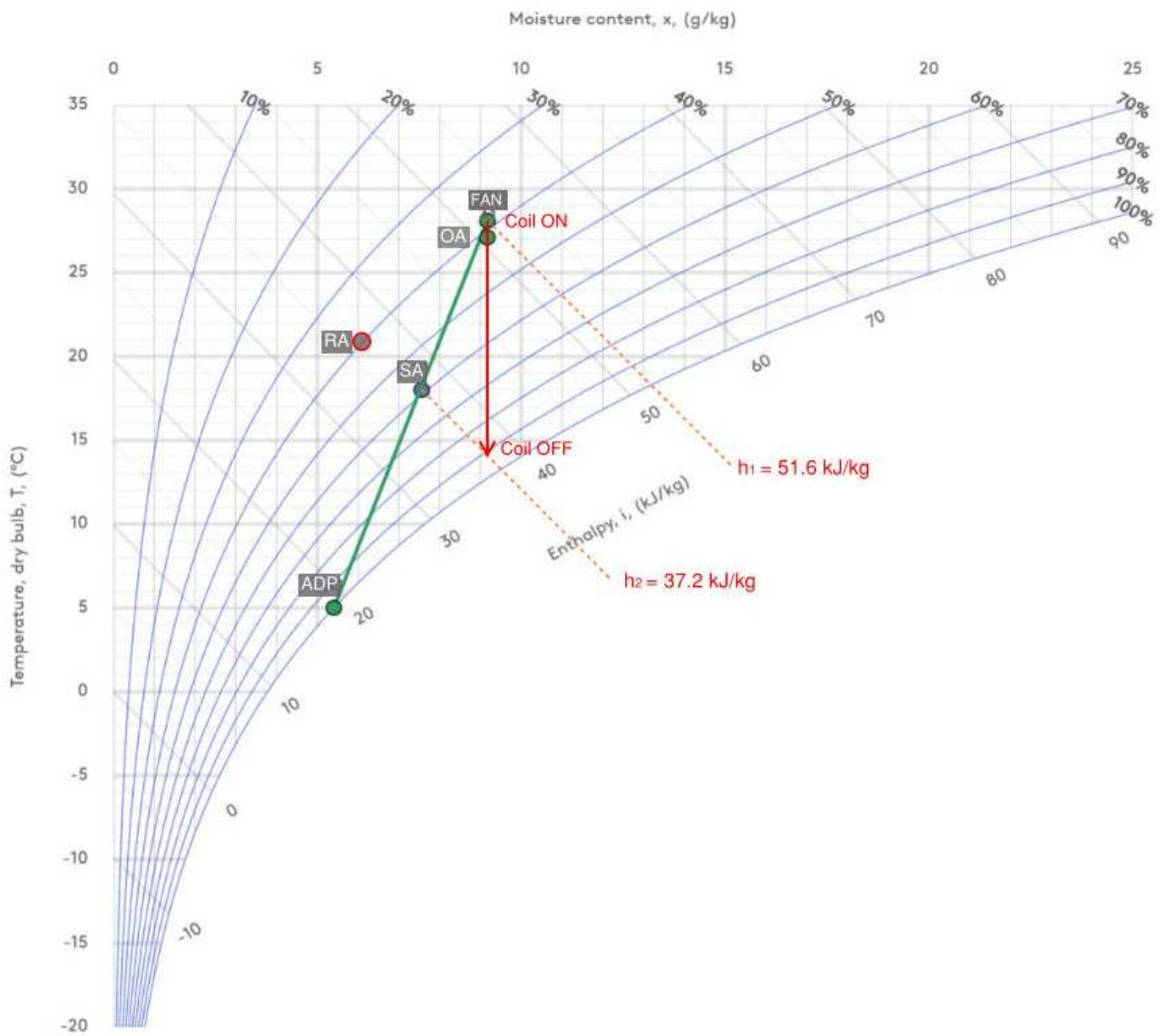


Figure 04 2 - AHU cooling coil dimensioning showing Outdoor air (OA) cooled to Supply air (SA) following the direction to ADP